



LONGEVITY & ANTI-AGING

Pinealon

Pinealon – 20 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

|                     |                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| CHEMICAL NAME       | L-Glutamyl-L-aspartyl-L-arginine                                                                                                           |
| SYNONYMS / ALIASES  | Pinealon; EDR tripeptide                                                                                                                   |
| CAS NUMBER          | Not formally registered in standard chemical registries (Russian-research peptide; see published references for structural identification) |
| MOLECULAR FORMULA   | C15H26N6O8                                                                                                                                 |
| MOLECULAR WEIGHT    | Approximately 418.4 g/mol                                                                                                                  |
| AMINO ACID SEQUENCE | EDR (3 residues; H-Glu-Asp-Arg-OH)                                                                                                         |
| SOURCE              | Synthetic (solid-phase peptide synthesis); designed as a short-peptide "bioregulator" analog within the Khavinson peptide framework        |

PHYSICAL PROPERTIES

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| APPEARANCE           | Lyophilized white to off-white powder                               |
| PURITY SPECIFICATION | >=99% by HPLC (confirm against batch COA)                           |
| CONTAINER            | 20 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp |

STORAGE & HANDLING

|                                      |                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------|
| UNRECONSTITUTED STORAGE              | -20 C, protect from light and moisture                                        |
| RECONSTITUTED STORAGE                | 2-8 C, stable approximately 28 days in bacteriostatic water                   |
| SUGGESTED RECONSTITUTION SOLVENT     | Bacteriostatic water or sterile water (research use only)                     |
| SHELF LIFE (UNRECONSTITUTED, SEALED) | 24 months at -20 C                                                            |
| STABILITY NOTES                      | Small linear tripeptide; generally stable. Avoid repeated freeze-thaw cycles. |

## STRUCTURAL NOTES

Pinealon is a synthetic tripeptide (3 amino acids) of sequence Glu-Asp-Arg. It belongs to the Russian-research "Khavinson peptide" family of short regulatory peptides (typically tri- or tetrapeptides) characterized in geriatric and neuroscience research literature. Within that family, Pinealon is grouped with peptides characterized for activity in nervous-system research models.

## SELECTED PEER-REVIEWED REFERENCES

1. Khavinson V.K., Malinin V.V. - *Gerontological Aspects of Genome Peptide Regulation* - Karger, 2005 - ISBN: 3-8055-7825-9
2. Khavinson V.K. et al. - *Tripeptides slow down the aging process in cell cultures of human skin fibroblasts* - Bulletin of Experimental Biology and Medicine, 2011 - DOI: 10.1007/s10517-011-1377-x
3. Arutjunyan A. et al. - *Pinealon protects the rat offspring from prenatal hyperhomocysteinemia* - International Journal of Clinical and Experimental Medicine, 2012 - PMID: 22808348

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

## COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.